

Research Notes: Variance, the Law of Large Numbers, and the Central Limit Theorem (Part-2)

Chebyshev's Inequality · From First Principles · A Complete Derivation

Log

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Methodology: the user drives every step of the derivation; the assistant only executes computations and judges correctness. Core principle: no presupposed conclusions — the record of trial and error is itself part of the manuscript.

[Entry 001] Statement and historical position

Statement: for any random variable X (only the mean μ and variance σ^2 need exist) and any $k > 0$:

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

The probability of deviating from the mean by more than k is capped by the variance over k^2 . No knowledge of the distribution's shape is needed — it holds for *every* distribution, which is its entire value; the price is a loose bound (a worst-case bound: the tail is only pushed down to $O(1/k^2)$ polynomial decay).

Historical position: the CLT came first.

- 1713, Bernoulli: the first LLN (Ars Conjectandi), brute-force tail estimates specific to the binomial; no general-purpose inequality.
- 1733, de Moivre: the binomial CLT (the path of Part-1).
- 1812, Laplace: generalization to arbitrary p .

- 1853 / 1867, Bienaymé / Chebyshev: this inequality appears. Chebyshev’s motivation was precisely to free the LLN from the binomial — every earlier LLN proof was chained to distribution-specific combinatorial estimates.

Logical strength and historical order run in opposite directions: Chebyshev uses only the second moment, is elementary and non-asymptotic (valid for every finite n), but loose; the CLT is a strictly finer statement — it gives the *shape* of the fluctuations at scale $\sigma\sqrt{n}$ (Gaussian tails, $e^{-x^2/2}$ -level decay). The modern textbook order (Chebyshev → LLN first, CLT after) is sorted by logical difficulty, not history. The inequality has zero dependence on the 1812 generalization — fully self-contained, using nothing but the single definition “variance = expected squared deviation.”

[Entry 002] Opening question: must the integral and discrete cases be proved separately?

Q: it doesn’t yield to a direct attack — should the continuous and discrete cases be proved separately?

A: no. Write it in expectation form, $\sigma^2 = E[(X - \mu)^2]$; the single symbol E covers both discrete sums and continuous integrals, and the proof is verbatim identical for both. Below we write the continuous case (assuming a density $p(x)$ exists); the discrete case follows word for word with integrals replaced by sums.

[Entry 003] Dead end one: expanding by definition

The user’s first attempt:

$$\sigma^2 = E[(X - \mu)^2] = E(X^2) - \mu^2 = \int_{-\infty}^{\mu-k} p(x)x^2 dx + \int_{\mu-k}^{\mu+k} p(x)x^2 dx + \int_{\mu+k}^{\infty} p(x)x^2 dx - \mu^2$$

Every equality holds. But the user immediately pronounced it dead: $P(|X - \mu| \geq k)$ **cannot be assembled from this**. Correct verdict — the step $E(X^2) - \mu^2$ shatters the structure of $(x - \mu)^2$, and the target event cannot be reassembled from the fragments x^2 and μ^2 . Retreat.

[Entry 004] Wrong diagnosis: blaming the “order”

The user proposed: “we need to conjure a first-order term; second order won’t work.”

Verdict: no — both halves fail. No first-order term is needed, and second order is not the problem. What died was the **expansion** in the previous step, not the order of the moment. (Wrong diagnosis, but the instinct to change tactics was right.)

[Entry 005] The key step: keep $(x - \mu)^2$ intact, cut the domain by the event

Q (yes/no format): “ $(x - \mu)^2$ can’t be expanded, right?”

A: don’t expand it — not that it “can’t” be, but that in this proof it **must be kept whole**; expanding returns to the dead end of Entry 003.

Cut the integration domain by the event $|X - \mu| \geq k$ (this step supplied by the assistant at the user’s explicit instruction “you derive one step”):

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx = \underbrace{\int_{|x-\mu| < k} (x - \mu)^2 p(x) dx}_{\text{first term (good event)}} + \underbrace{\int_{|x-\mu| \geq k} (x - \mu)^2 p(x) dx}_{\text{second term (bad event)}}$$

The cutting principle: split the domain by the **target event**, not by algebraic intervals of x — the event itself must appear in the formula as a single whole.

[Entry 006] Dead end two: bounding in the wrong direction

The user’s second attempt: “the first term can be bounded — $\sigma^2 \leq k^2 + \text{second term}$.”

Verdict: the inequality itself is true (first term $\leq k^2 \cdot P(|X - \mu| < k) \leq k^2$), but as a proof step it is dead — this is an **upper** bound on σ^2 , and no upper bound on $P(|X - \mu| \geq k)$ can be extracted from $\sigma^2 \leq k^2 + \text{second term}$; the first term’s information is slack in the wrong direction. The bounding direction is inverted.

[Entry 007] The correct bound; QED

The user flipped the direction: “ $\sigma^2 \geq \text{first term} + k^2 \int p(x) dx$ — and it falls out.”

Verdict: correct. Two bounds:

1. On the bad-event domain, **pointwise** $(x - \mu)^2 \geq k^2$, so the second term $\geq k^2 \int_{|x-\mu| \geq k} p(x) dx$;
2. Throw away the good-event domain wholesale: first term ≥ 0 .

$$\sigma^2 \geq k^2 \int_{|x-\mu| \geq k} p(x) dx = k^2 P(|X - \mu| \geq k)$$

Rearranging gives $P(|X - \mu| \geq k) \leq \sigma^2/k^2$. QED.

Three-move structure: cut the domain (keeping $(x - \mu)^2$ intact) $\rightarrow$ pointwise lower bound k^2 on the bad event $\rightarrow$ discard the good event wholesale. The cause of death in Entry 003 also becomes clear: $E(X^2) - \mu^2$ destroyed the very object $(x - \mu)^2$ on which the pointwise lower bound rests.

| [Entry 008] Pitfalls (4 total)

Pitfall 1: expanding by definition to $E(X^2) - \mu^2$. The expansion shatters $(x - \mu)^2$; the pointwise lower bound loses its carrier, and the target event cannot be reassembled from fragments. Lesson: when proving tail bounds, “squared deviation” must be manipulated as a single object.

Pitfall 2: attributing the failure to “order”. “Need a first-order term; second order won’t work” — both halves wrong. What died was the expansion as an operation, not the order of the moment: the entire proof uses only the second moment.

Pitfall 3: bounding in the inverted direction. $\sigma^2 \leq k^2 + \text{second term}$ is a true statement but useless: the goal is an upper bound on the tail probability, equivalent to a **lower** bound on σ^2 ($\sigma^2 \geq k^2 P$); nothing can be extracted from the upper-bound direction. Before bounding, be explicit: what is being squeezed, and toward which side.

Pitfall 4 (collaboration error, non-mathematical): the assistant misattributed a quote. The assistant credited to the user an intuition sentence the assistant itself had said two turns earlier (“that line from your intuition earlier...”); the user caught it on the spot and the attribution was retracted. On record: in conversational derivation, the provenance of a claim is itself part of the ground truth.

| Appendix: the whole proof in three lines

$$\sigma^2 = \int_{|x-\mu| < k} (x - \mu)^2 p \, dx + \int_{|x-\mu| \geq k} (x - \mu)^2 p \, dx \geq 0 + k^2 \int_{|x-\mu| \geq k} p \, dx = k^2 P(|X - \mu| \geq k)$$

End